

USTHB University
Faculty of Computer Science
AI&DS Department

DevOps & Cloud Computing

03- Cloud Computing Architecture

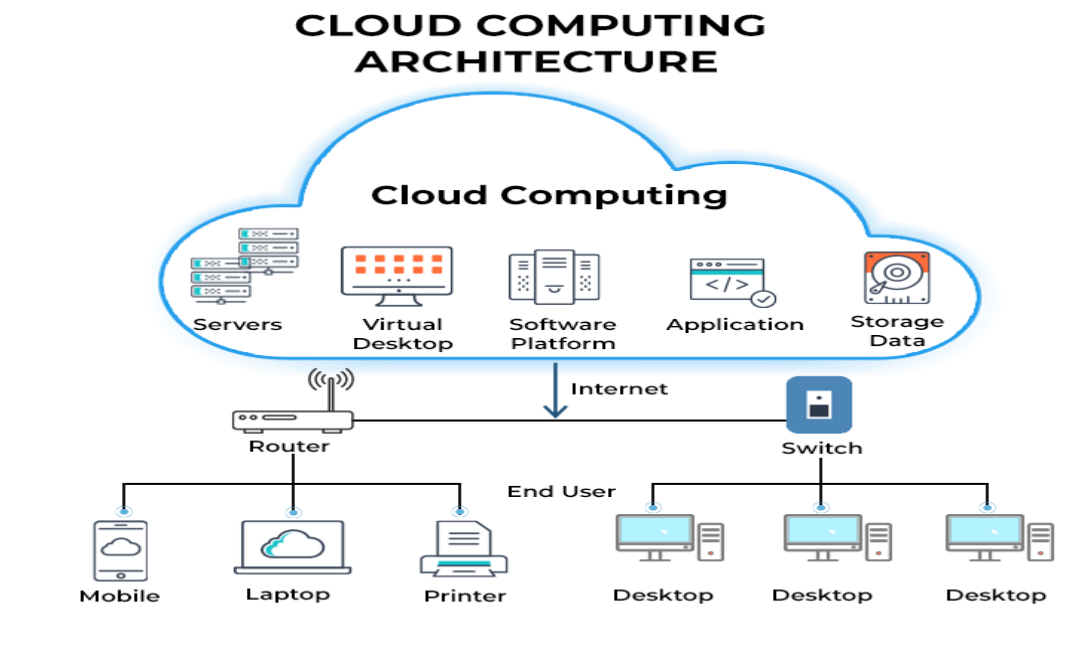






Engineers (5th Year) | AMEUR IMANE | 2026/2027

1. Cloud Computing Architecture



Cloud Computing Architecture

Resources can be adjusted according to needs (scaling up or down).

Cloud architecture refers to the **design** and **management of computing resources** and **services** that are based on cloud computing.

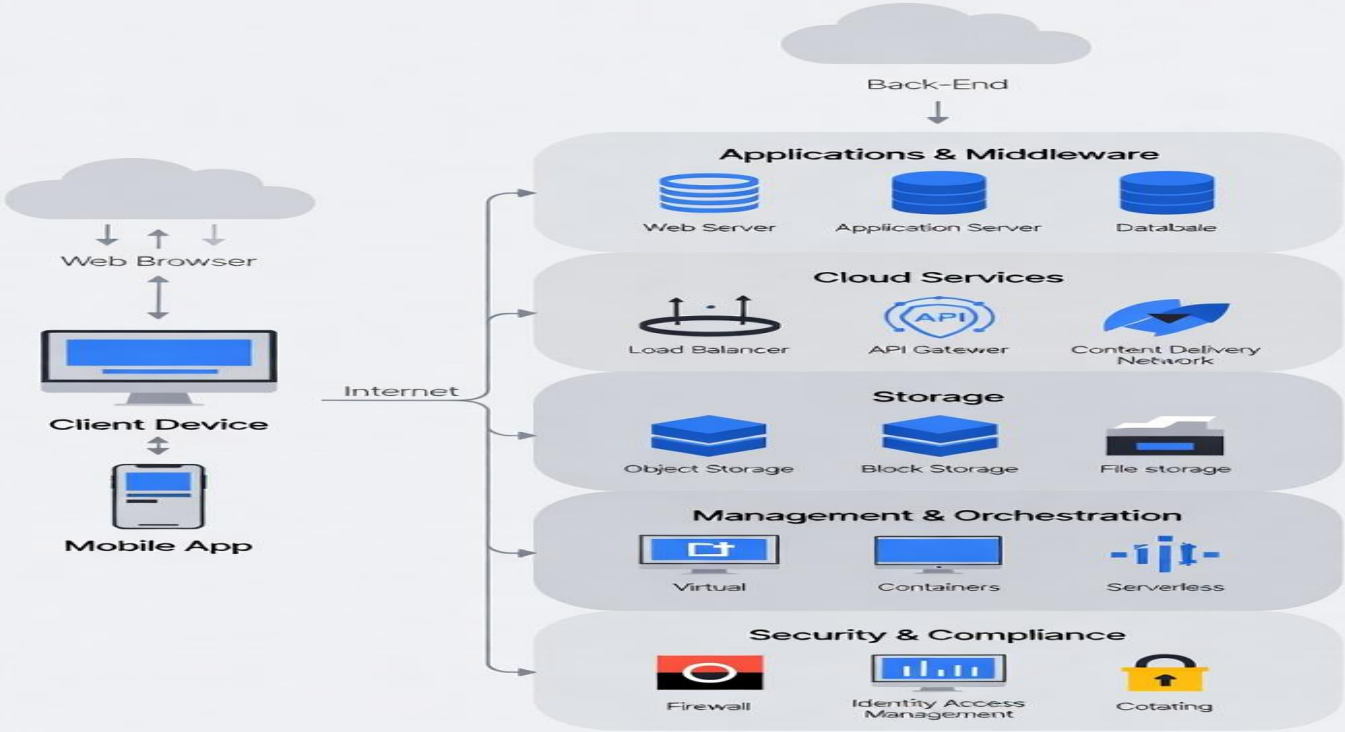
Cloud providers ensure high availability and fast response times through redundant infrastructures and fault-tolerance mechanisms.

It aims to provide **scalable, reliable, secure, and cost-effective solutions** for modern applications and businesses.

Providers implement advanced security protocols.

By eliminating the costs of physical infrastructure, cloud architecture allows companies to reduce their capital expenditures.

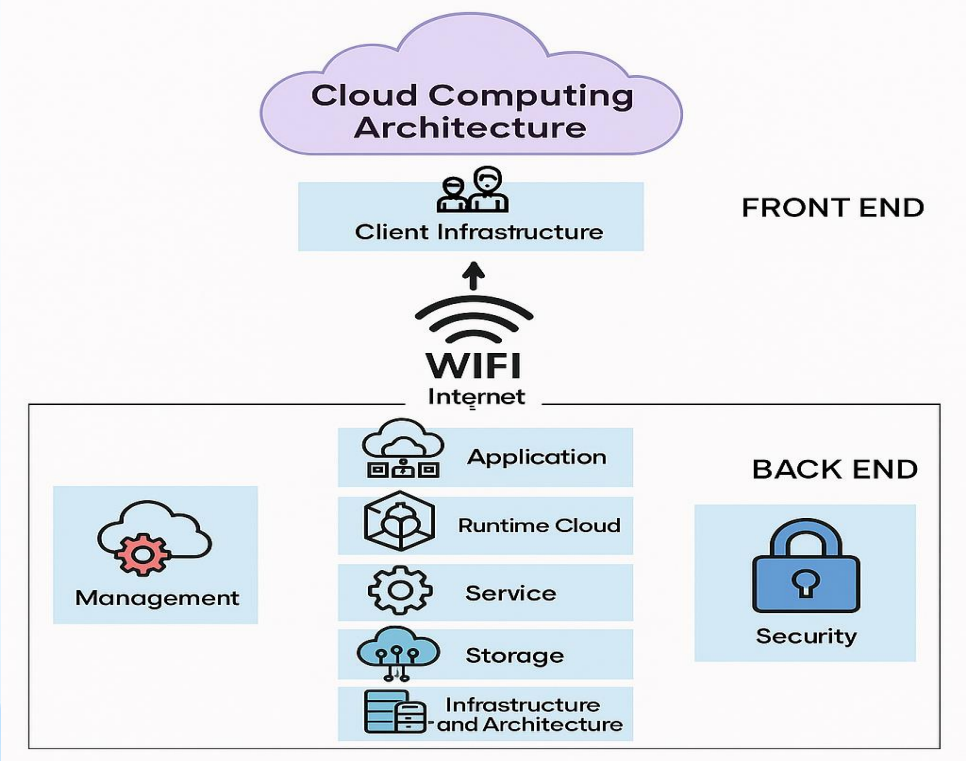
Cloud Computing Architecture



Cloud Computing Architecture

Cloud architecture consists of two main parts:

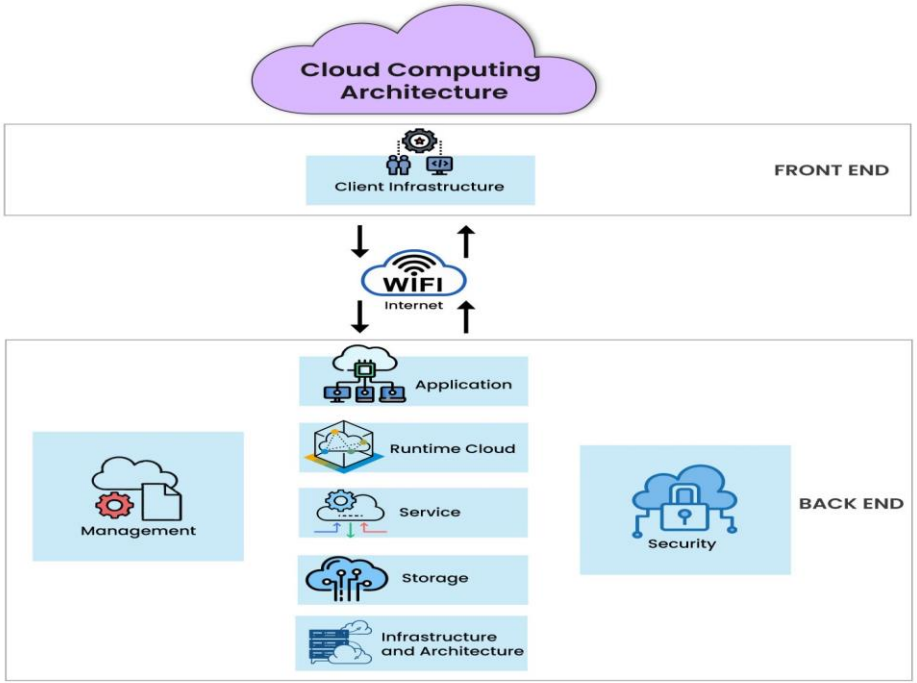
- **Front End**
- **Back End**



Cloud Computing Architecture

Introduction

- **Front-End:** Cloud front-end architecture refers to all the components used by the **end user** to interact with cloud computing services.
- It includes the user-side applications and interfaces required to access services hosted in the cloud.

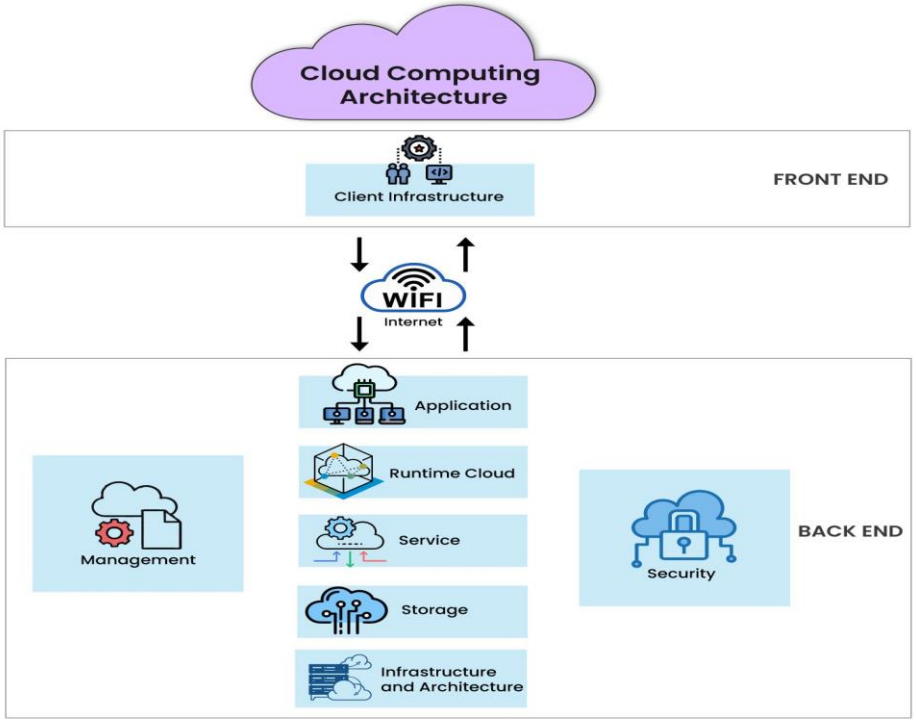


Cloud Computing Architecture

Front-End

The key elements of this architecture are the main components:

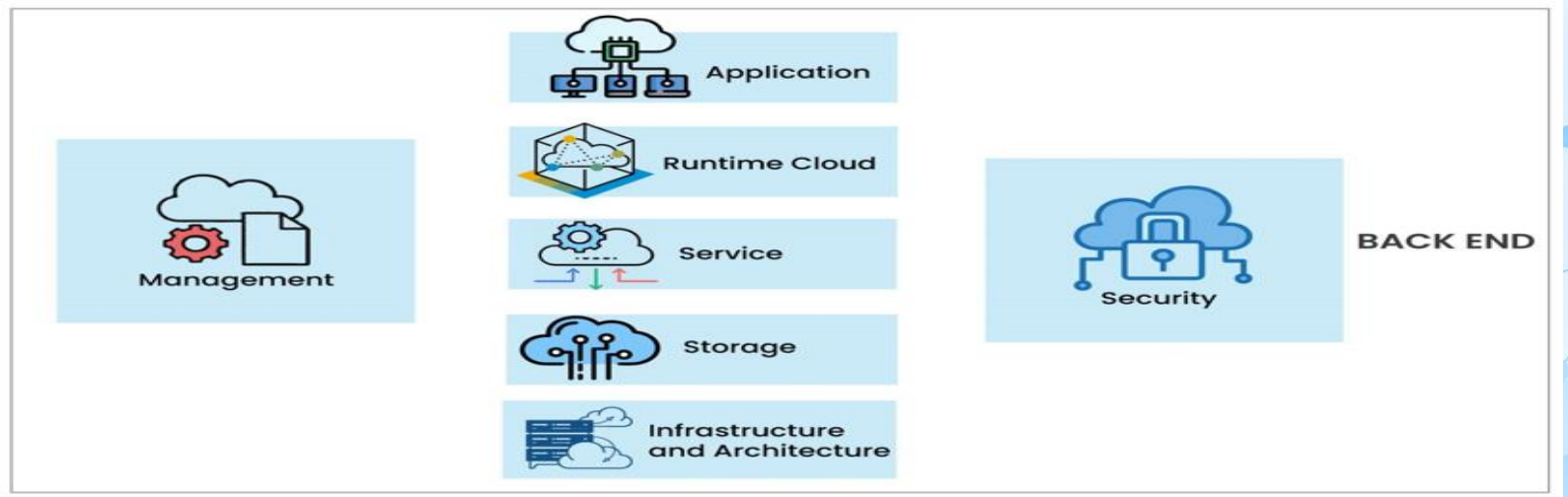
- **The Client Device**: Desktop computer, smartphone, tablet, etc.
- **The Web Browser (or dedicated application)** used to access the cloud interface.
- **The User Interface (UI) and client applications** (e.g., the cloud provider's web portal, or a SaaS application like Gmail or Salesforce).



Cloud Computing Architecture

Back-End

Is used by the cloud service provider, who manages and controls all the resources required to deliver cloud services. It is an essential element of the overall cloud computing architecture, as it enhances the functionality of the front-end while protecting cloud data.



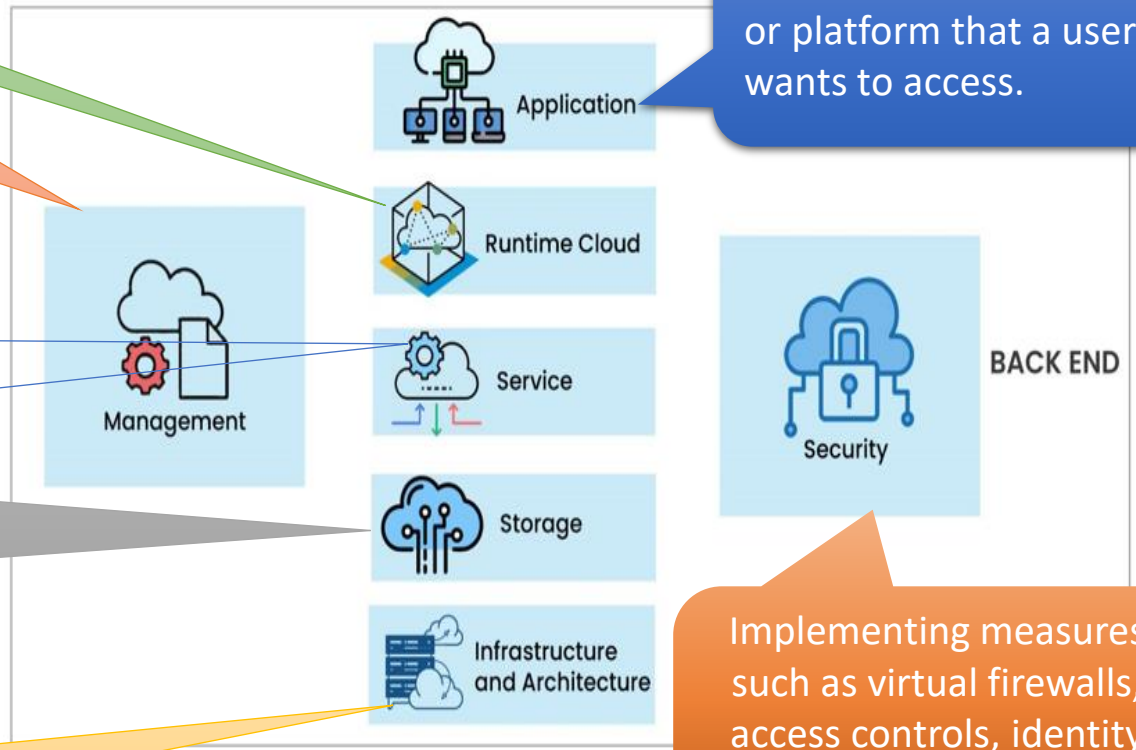
Cloud execution environment

Responsible for monitoring and managing back-end components

A component that manages every operation running on the cloud computing platform (SaaS, PaaS, IaaS)

Provides flexible and scalable cloud storage services (e.g., Microsoft Azure Storage, Amazon S3, Oracle Cloud Storage, etc.)

Composed of software and hardware resources such as network devices and servers



It can be any software or platform that a user wants to access.

Implementing measures such as virtual firewalls, access controls, identity management, and data encryption…

Cloud Computing Architecture

Back-End (Provider/Server Side)

Role: Manage physical and virtual infrastructure, storage, security, automation, and execute the business logic of applications.

Main Components:

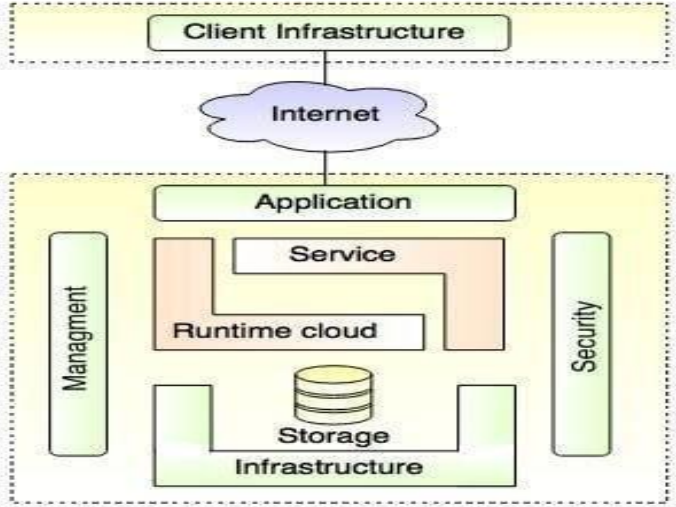
- **Physical Infrastructure:** Servers, storage, and network equipment (in the data center).
- **Management Software:** Hypervisors (such as VMware or KVM), and cloud management platforms (such as OpenStack).
- **Storage:** Databases and file systems for data storage.
- **Security and Backup Mechanisms:** Authentication protocols, firewalls, and redundancy/backup systems.

Cloud Computing Architecture

Cloud computing relies on a set of **architectural principles** that ensure **flexibility**, **scalability**, and **reliability**.

- Key components include **multi-tier structures**, **virtualization technologies**, **abstraction mechanisms** and **service-orientation**.

- These architectures allow **resources** and **services** to be delivered **on demand** over the internet.



Cloud Computing Architecture

A) N-Tiers

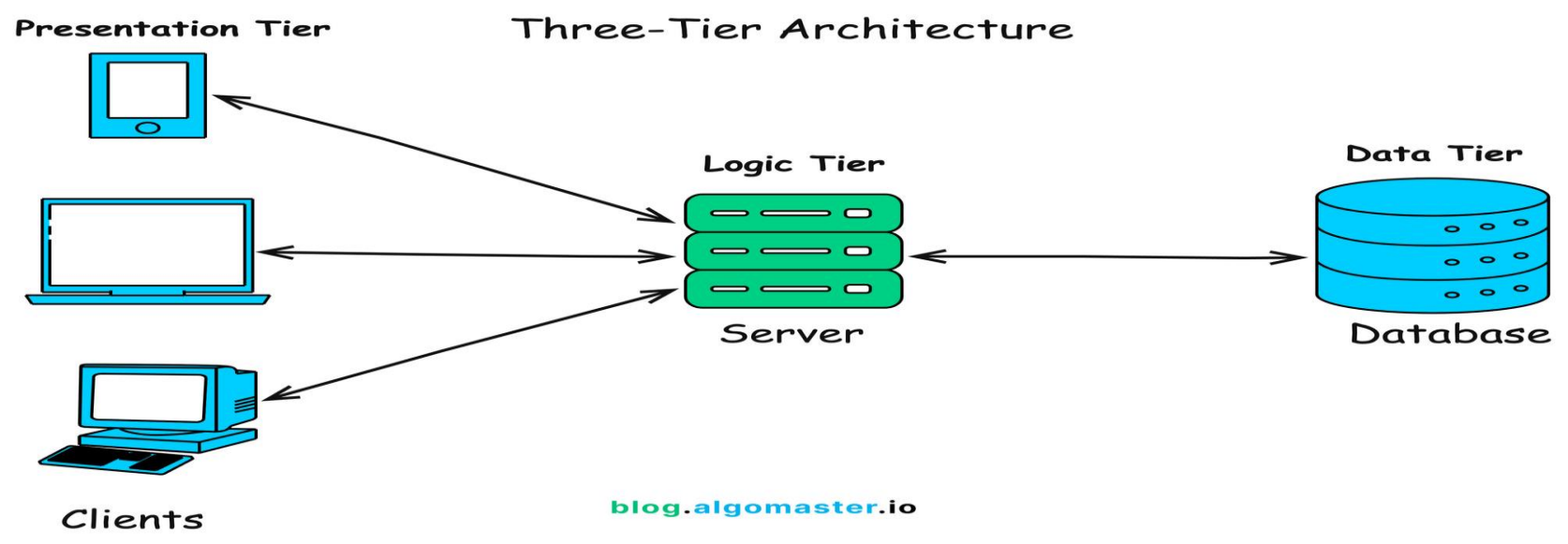
"**N-Tiers Architecture**" is a **Concept** in development and other areas.

1) Principle

- Software design pattern that divides an application into logical layers (tiers).
- Each tier handles a specific responsibility, promoting separation of concerns.
- Common Tiers:
  - Presentation (UI/UX).
  - Business Logic (rules, computations).
  - Data (database, storage).
- Improves *maintainability*, *scalability*, and *security*.

Cloud Computing Architecture

A) N-Tiers



Cloud Computing Architecture

A) N-Tiers

2) N-Tiers in Cloud Computing:

- Cloud-native applications adopt N-Tiers for distributed deployment.

- Example:

  - **Web Frontend** → hosted on S3 / CloudFront.
  - **Business Logic** → runs on Lambda, App Engine, or EC2.
  - **Data Layer** → managed by cloud databases like RDS, BigQuery.

- Benefits:

  - **Independent** scaling of each tier.
  - **Simplifies updates** and **fault isolation**.
  - **Enhances flexibility** for *hybrid* and *multi-cloud* setups.

Cloud Computing Architecture

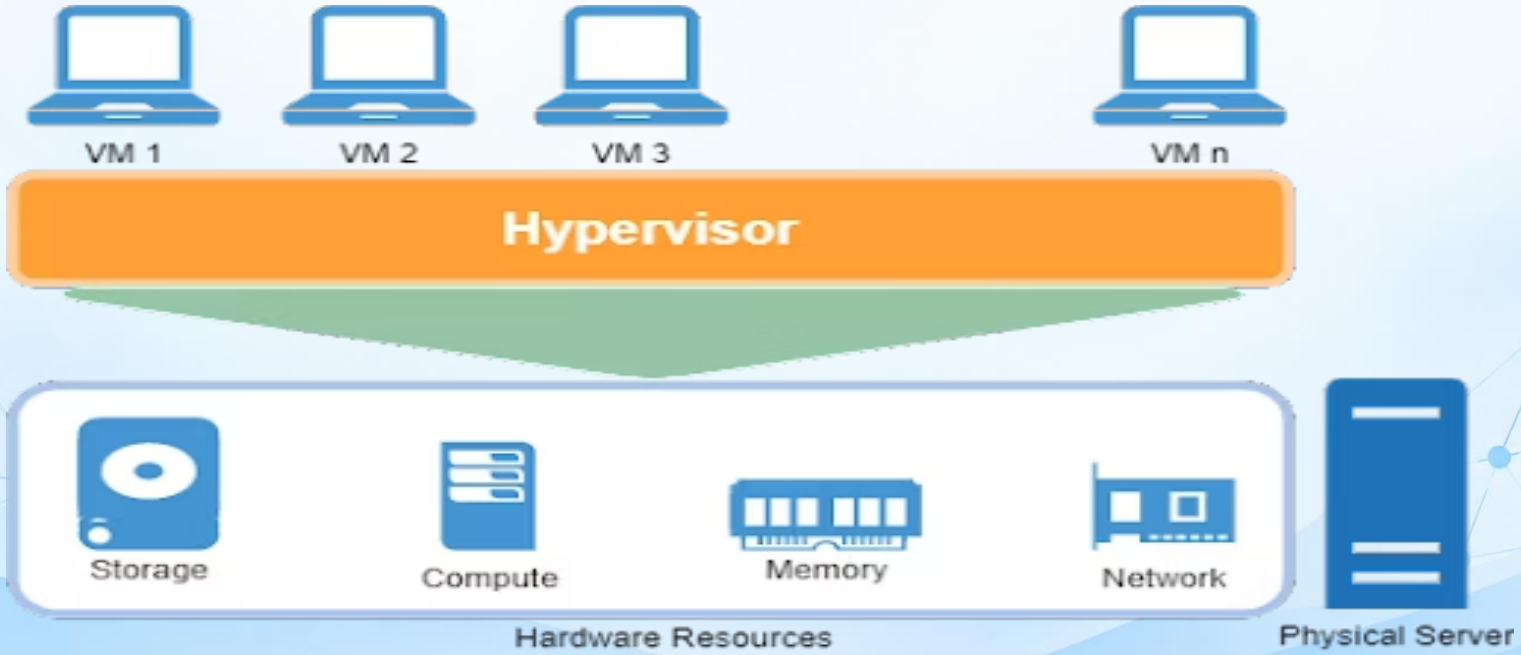
B) Virtual Machines (VMs)

"**Virtual Machines**" in simple words are Software **emulation** of **physical computers**.

- VM Components:
  - + **Host OS**: Manages hardware.
  - + **Hypervisor**: Manages VMs.
  - + **Guest OS**: Runs inside the VM.
- Offers **isolation**, **portability**, and **efficient resource use**.

Cloud Computing Architecture

B) Virtual Machines (VMs)



Cloud Computing Architecture



C) File Virtualization

"**File virtualization**" abstracts file storage from **physical devices** and **locations**.

1) Principle

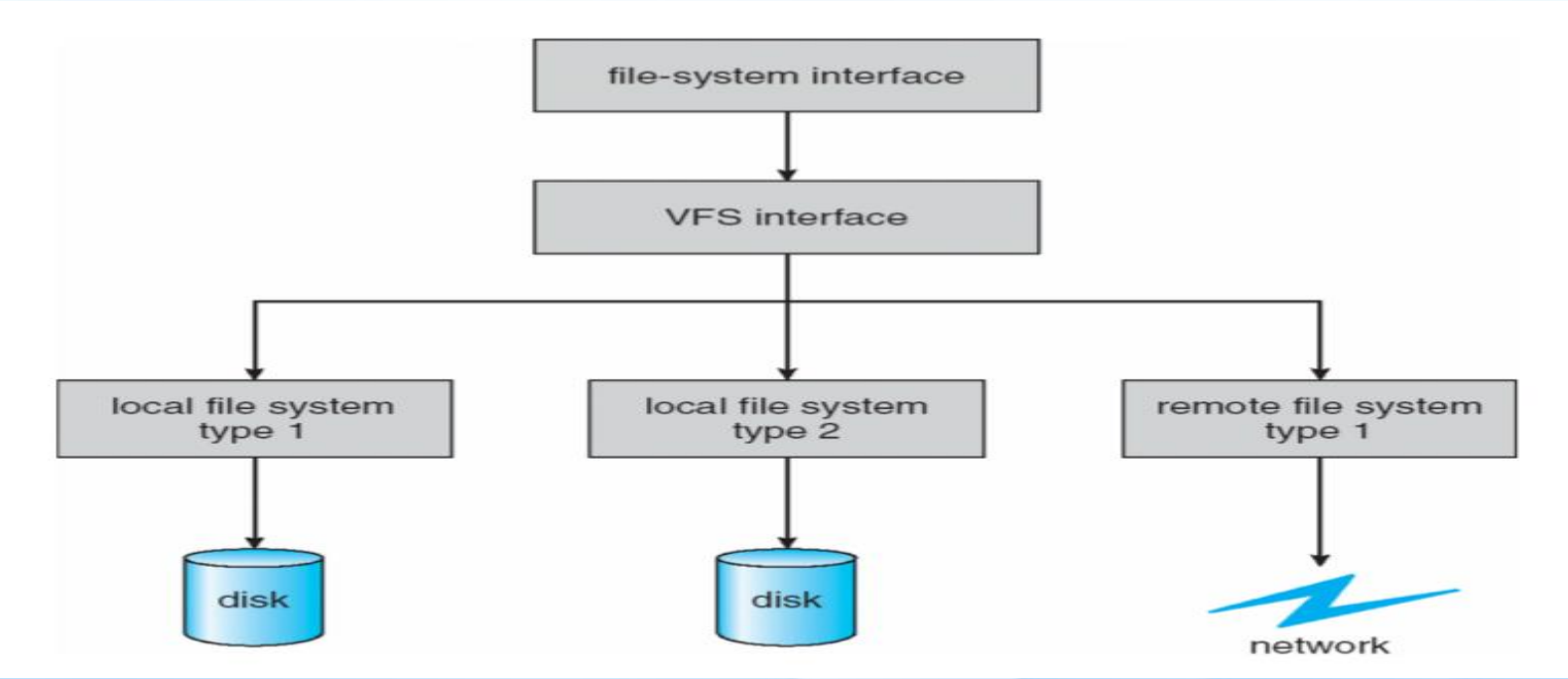
- Decouples file access from physical storage management.
- Provides a unified view of distributed file systems.

2) Technologies

- Virtual File Systems (VFS).
- Storage virtualization *appliances* or *software*.

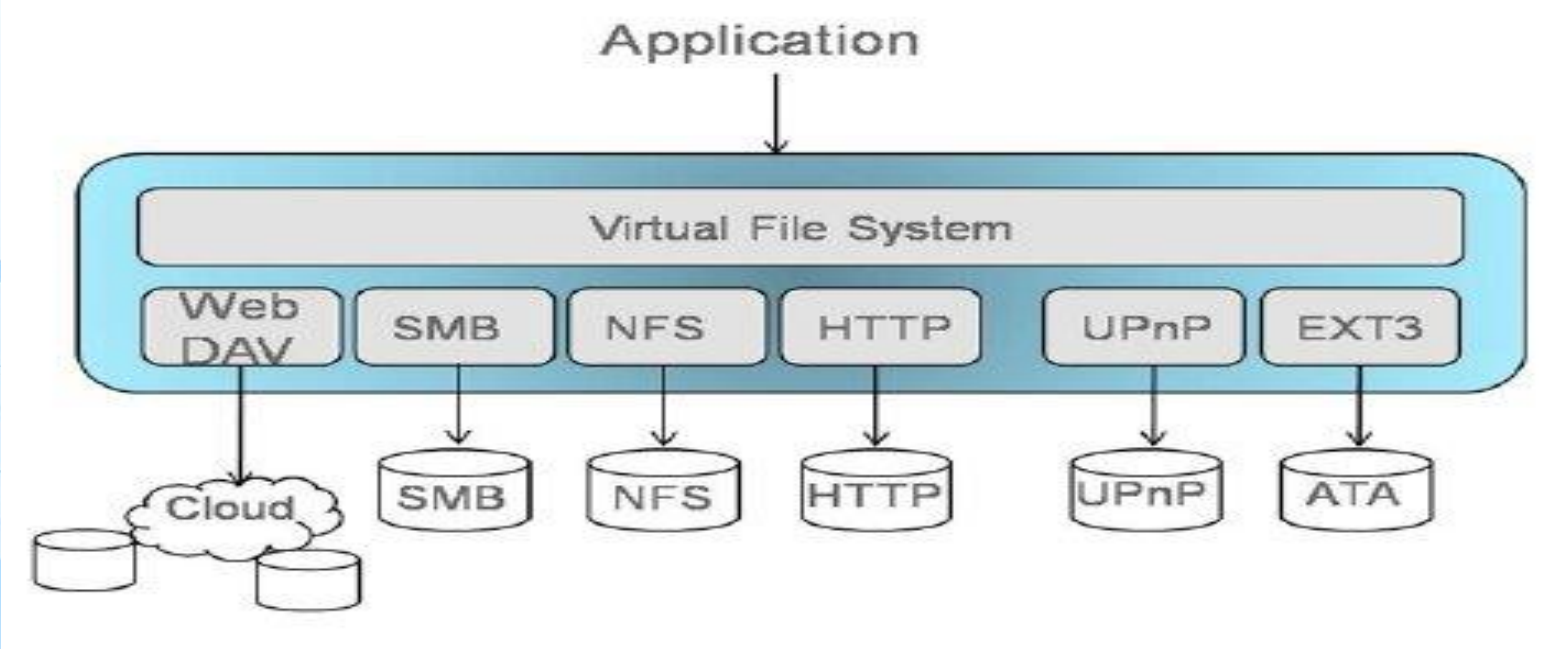
Cloud Computing Architecture

C) File Virtualization



Cloud Computing Architecture

C) File Virtualization



C) File Virtualization

3) Role in Cloud

- Used in distributed file systems and cloud storage solutions (e.g., *Dropbox*, *Google Drive*, *S3*, etc.).

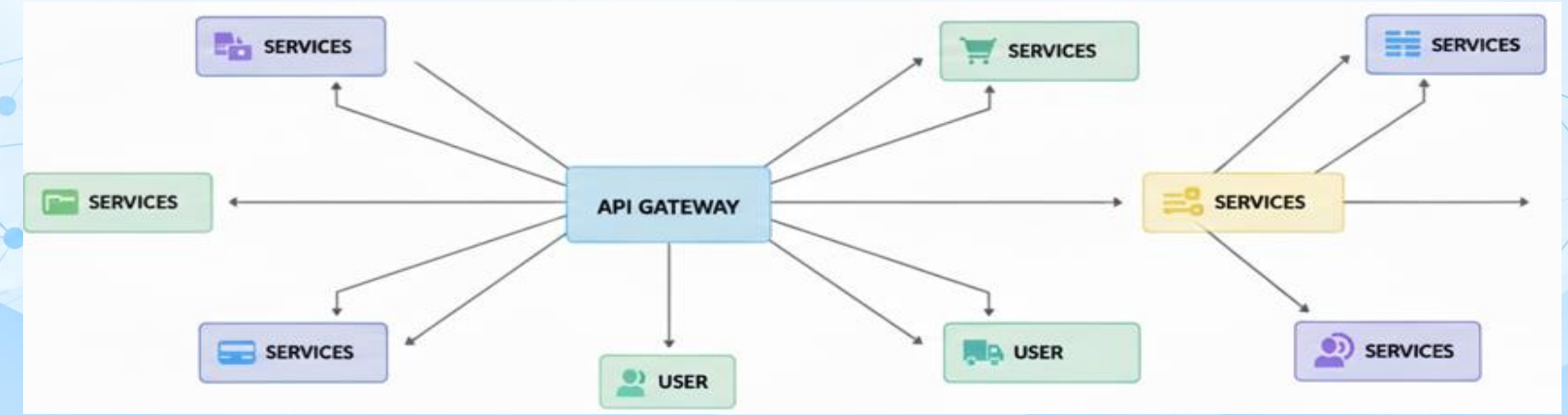


Cloud Computing Architecture

What is Service-Oriented Architecture (SOA)?

D) Service-Oriented Architecture (SOA)

Service-Oriented Architecture (SOA) consists of designing an application as a set of **independent services**, each performing a specific business function and communicating via **APIs** (Application Programming Interfaces).



Cloud Computing Architecture

D) Service-Oriented Architecture (SOA)

"SOA Architecture" is a style where software components are designed as **loosely coupled** and **interoperable services** that communicate over a network.

1) Principle

- SOA is an architecture where services are:
  - Loosely coupled.
  - Discoverable and reusable.
  - Communicate via standardized protocols (**SOAP**, **REST**).
- Encourages **flexibility** and **modularity**.
- Services are **platform-independent** and **interoperable**.

D) Service-Oriented Architecture (SOA)

Main characteristics

Characteristics	Description
Loose coupling	Each service is independent from the others.
API-based communication	REST, gRPC, GraphQL, SOAP, etc.
Reusability	Services can be reused by multiple applications.
Technological independence	Each service can be developed using a different programming language.

REST	**Re**presentational **S**tate **T**ransfer
gRPC	**g** (Google) **R**emote **P**rocedure **C**all
GraphQL	**Graph Q**uery **L**anguage
SOAP	**S**imple **O**bject **A**ccess **P**rotocol

Cloud Computing Architecture

D) Service-Oriented Architecture (SOA)

2) SOA in Cloud Computing

- Backbone of cloud-native and hybrid systems.
- Cloud platforms expose services via **APIs** (e.g., *AWS Lambda*, *Azure Functions*).

- It Benefits:
  - **Dynamic** service **discovery** and **binding**.
  - **Scalability** and **reusability** across cloud ecosystems.
  - **Smooth integration** in multi-cloud strategies.

SOA Advantages

Advantage	Explanation
Fast deployment	Containers start in a few seconds.
Cost efficiency	Only the necessary services are scaled.
Resilience	A failure in one service does not stop the application.
Technological flexibility	Each service can use the best suitable technology.
Continuous updates	A single service can be updated without affecting the rest.

Cloud Computing Architecture

Real-world use cases

Domain	Examples
E-commerce	(ex: Amazon, Jumia, Alibaba) : Cart, payment, search, delivery
Banking	Payments, fraud detection, customer management
Transport	Uber-like : Booking, pricing, geolocation
Healthcare	Medical records, billing, consultations, laboratories

Cloud Computing Architecture

Example: An E-commerce platform (Amazon, Jumia, Alibaba)

- **User Service** handles authentication (login/signup).
- **Product Service** manages product listings.
- **Order Service** handles orders and payments.
- **Notification Service** sends emails or SMS.

- Each service **runs independently**, **communicates via APIs**, can be **deployed** across **cloud** or **hybrid** environments.
- When a user places an order, the **Order Service** talks to **User Service** (*verify user*), **Product Service** (*check stock*), and **Notification Service** (*confirm order*); all **modular**, **scalable**, and **replaceable**.

Cloud Computing Architecture

From SOA to Serverless: Serverless Computing

“Deploy code without managing servers”

“Serverless Computing” is a cloud model in which developers run **code** without managing server infrastructure, virtual machines, containers, or operating systems.

The cloud automatically takes care of: creating the necessary servers, running the code, scaling according to demand, stopping resources, and charging only for what is actually used. The developer, on the other hand, only focuses on the ***code*** and the ***events that trigger its execution***.

Cloud Computing Architecture

From SOA to Serverless: Serverless Computing

Operation (Function as a Service - FaaS): The most common serverless model is known as **FaaS**. Developers write pieces of **code** called **functions** (**such as AWS Lambda or Google Cloud Functions**) that are executed only in response to an event (for example, a user clicks a button or a file is uploaded).

Examples of events:

- A user uploads an image to cloud storage.
- A change is made in a database.
- An HTTP request (API call) is received.
- A message arrives in a queue.

Cloud Computing Architecture

Payment Model: Consumption-Based

Pay-as-you-go: A major advantage is the pricing model. Unlike traditional servers where you pay for **the time the server is running**, with **serverless** you only pay for **the execution time of your code** (often measured in milliseconds).

Cloud Computing Architecture

Main Serverless Services

Cloud	*Service Serverless*	*Description*
AWS	Lambda	The wellknown one
Azure	Azure Functions	Event-driven Fonctions
Google Cloud	Cloud Functions / Cloud Run	Cloud Run supports containers
IBM Cloud	IBM Cloud Functions	Based on Apache OpenWhisk
Open-source	OpenFaaS, OpenWhisk, Knative	For private Cloud or on-prem

Advantages of Serverless

- **No infrastructure management**: Developers do not need to manage servers.
- **Automatic scaling**: Resources scale automatically based on demand.
- **Cost efficiency**: You pay only for actual execution time.
- **High availability**: Built-in fault tolerance and reliability.
- **Faster development**: Focus is only on writing code, not managing infrastructure.
- **High performance:** the platform automatically allocates resources based on workload

Tutorial exercises: 1

A company wants to develop an online shopping platform that must support 50,000 simultaneous users. The platform includes a web interface, order processing, payment services, and a database.

1. Propose a suitable cloud architecture by identifying the Front End, Back End, Business Logic, and Data Layer.
2. Which cloud services would you use to store product images and customer information? Justify your choices.
3. Explain how the architecture can remain available if one server fails.
4. Should the company use a monolithic or microservices architecture? Justify your answer.

Tutorial exercises: 2

A university owns one powerful physical server but needs to provide separate environments for three departments: Computer Science, Mathematics, and Electronics.

1. Explain how virtualization can solve this problem.
2. How many virtual machines would you propose, and why?
3. What is the role of the hypervisor in this architecture?
4. If one VM consumes too much CPU, how could this affect the other VMs?
5. Explain the difference between VM migration and VM replication.

Tutorial exercises: 3

A hospital uses a cloud platform to store medical documents, patient images, and administrative files. Different computers running Windows and Linux must access these files.

1. Which storage solution would you recommend for sharing files between different operating systems? Justify your answer.
2. Explain the roles of SMB and NFS in this architecture.
3. What is the role of a Virtual File System (VFS)?
4. Should the hospital use object storage or block storage for medical images? Justify your answer.
5. Propose two mechanisms to protect stored data against unauthorized access and data loss.

Tutorial exercises: 4

A photo-sharing application allows users to upload thousands of images every hour.

1. Explain how AWS Lambda can automatically process uploaded images.
2. Identify the event that triggers the function.
3. Explain why FaaS is suitable for this application.
4. Compare the Pay-as-you-go model of serverless computing with traditional server hosting.
5. Identify one limitation of serverless computing and propose a solution.

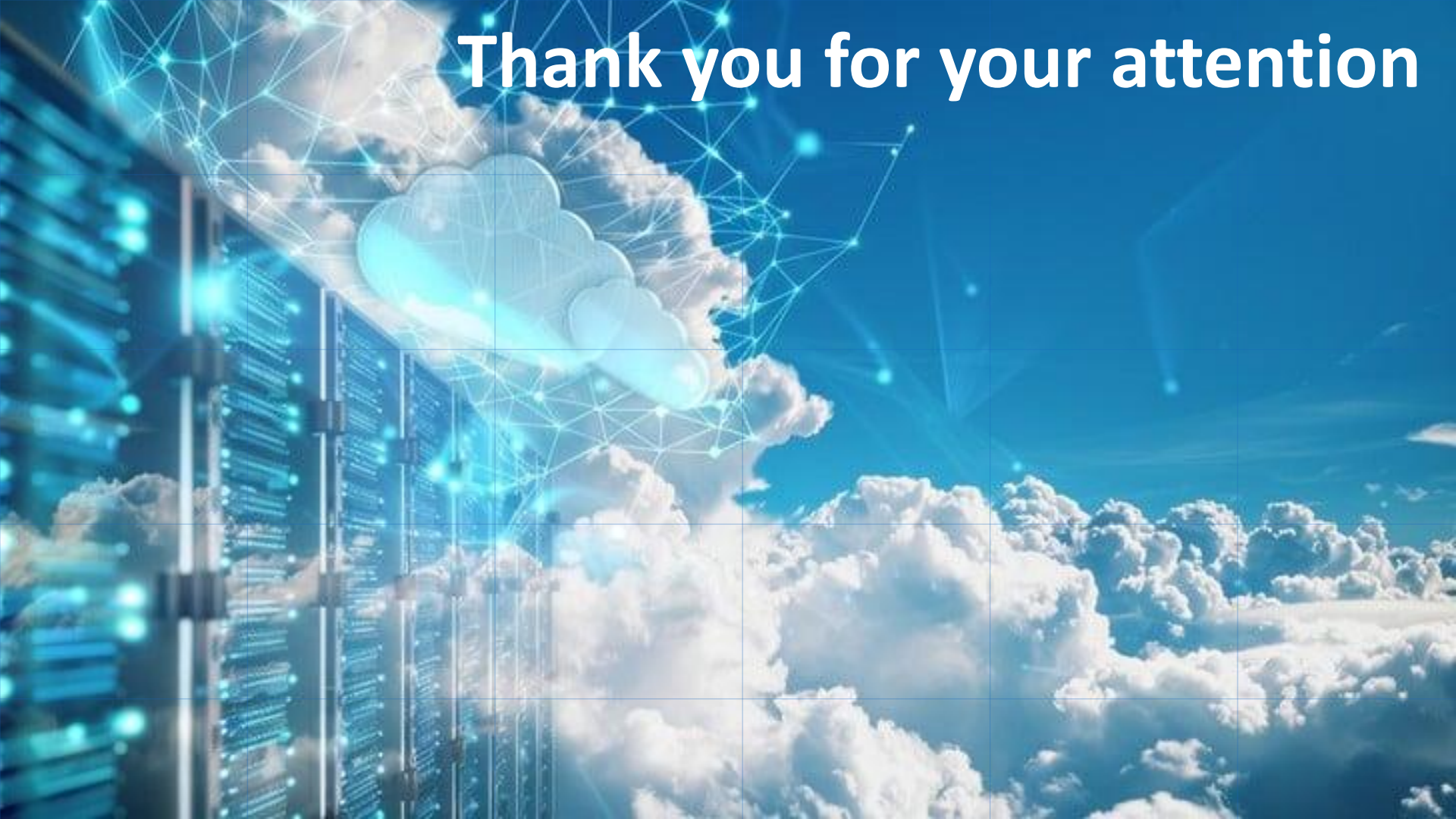
Tutorial exercises: 5

A bank develops an application using three independent services: customer management (Java), payment processing (Python), and notification services (Node.js).

1. Explain why this architecture is considered Service-Oriented Architecture (SOA).
2. Why are these services described as loosely coupled?
3. Explain how interoperability allows these services to communicate despite using different programming languages.

4. Compare SOAP and REST for communication between these services.

5. What happens if the notification service becomes unavailable? Does it necessarily stop the payment service? Justify your answer.


Thank you for your attention